CAPSTONE PROJECT REVIEW PROGRESS RECORD

Review: 02
Date of Review: 03.10.2026
Project Title: Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators
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Brief Description of the Project

Convolutional Neural Network (CNN) edge accelerators are constrained by the high area and power dissipation of Multiply-Accumulate (MAC) datapaths, where 8x8 exact multipliers dominate silicon cost (68 LUTs on Xilinx Zynq-7020). While approximate computing trades controlled arithmetic error for hardware savings, uniform approximation causes severe accuracy degradation or negligible gains. Existing layer-wise mapping methods rely on computationally prohibitive evolutionary search (ALWANN: 7.5 CPU-days) or GPU retraining loops (TRAM). This project proposes a lightweight, single-pass sensitivity metric computed in 0.2 seconds combining static weight variance and dynamic loss gradients via a 256-sample calibration probe without retraining. Layers are ranked such that error-vulnerable early layers receive exact multipliers while noise-resilient deep layers receive aggressive approximate multipliers. The heterogeneous design is mapped onto synthesizable Verilog Processing Elements (PEs) and deployed on a PYNQ-Z2 FPGA board via AXI4-Stream DMA, achieving up to 32.5% multiplier area reduction while preserving baseline classification accuracy.

Progress Achieved up to This Review

- **Baseline CNN & INT8 Quantization:** Trained a 4-layer CNN on Fashion-MNIST achieving 84.69% baseline test accuracy; implemented symmetric INT8 post-training quantization with negligible accuracy loss.
- **Single-Pass Sensitivity Probing:** Implemented TF GradientTape probe calculating sensitivity scores across a 256-sample calibration batch in 0.20s without retraining; proved conv1 is most sensitive (S=1.000) while conv4 is noise-resilient (S=0.014).
- **Validation Sweep & Signal Amplification:** Discovered mild truncation (T=4) operated within ±0.1% noise floor; scaled to aggressive T=6 (dropping 21 of 64 PPs, 6.6x NMED increase, 32.5% area savings) to expose true layer resilience differences.
- **Proof-of-Concept Accuracy Recovery:** Selective mapping with T=6 on conv4 recovered 81.5% of accuracy drop (85.27% accuracy) compared to uniform T=6 (84.00%), confirming the layer-wise hypothesis.
- **Spearman Correlation Gate Clearance:** Proposed metric S3 achieved Spearman rank correlation $|\rho| = 0.80$ on TRUNC=6, successfully clearing the mandatory $|\rho| >= 0.70$ GO gate and finalizing layer multiplier assignments.

Signature of Project Guide: _____ **Date:** _____